

INFRASTRUCTURE AUDIT REPORT

PROJECT : NETWORK SENTINEL | Date : April 27,2026

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-- PROJECT: NETWORK SENTINEL | AUDIT-ID: 0001
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TECHNICAL AUDIT : NETWORK INFRASTRUCTURE & AUTOMATED OBSERVABILITY

A Case Study in Incident Detection and Remediation

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Engineer :Mr.Kshitij Khaire
System   :Pop!_OS 24.04 LTS (COSMIC)
Kernel   :6.8.0
```

```

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```

kshitij@pop-os

```

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OS: Pop!_OS 24.04 LTS x86_64
Host: HP Pavilion 15 Notebook PC (088D10000030591)
Kernel: Linux 6.18.7-76061807-generic
Uptime: 4 hours, 56 mins
Packages: 2196 (dpkg), 21 (flatpak-user)
Shell: zsh 5.9
Display (LEN40B0): 1366x768 in 15", 60 Hz [Built-]
DE: COSMIC 1.0.0
WM: cosmic-comp (Wayland)
Theme: Fusion [Qt], Yaru [GTK3]
Icons: breeze-dark [Qt], Yaru [GTK3]
Cursor: Adwaita
Terminal: cosmic-term 1.0.9
Terminal Font: JetBrainsMonoNL Nerd Font Mono (12)
CPU: Intel(R) Core(TM) i7-3632QM (8) @ 3.20 GHz
GPU: Intel 3rd Gen Core processor Graphics Contro]
Memory: 3.52 GiB / 7.65 GiB (46%)
Swap: 0 B / 11.65 GiB (0%)
Disk (/): 65.77 GiB / 224.79 GiB (29%) - ext4
Disk (/recovery): 2.89 GiB / 3.99 GiB (72%) - vfat
Local IP (wlo1): 192.168.43.19/24
Battery (Primary): 63% [AC Connected]
Locale: en_US.UTF-8

```

Fig 1.1: Verification of the local staging environment and system specifications.

2.0 TECHNICAL IMPLEMENTATION : SCRIPT ARCHITECTURE

Establishing the 'Net_Sentinel' Source Code and Logic Flow

Source Code :

```
1 #!/bin/bash
2
3 # ---- THE NETWORK SENTINEL ----
4 # Purpose : Monitor Infrastructure Connectivity
5
6 TARGET="8.8.8.8"
7 DOMAIN="google.com"
8 LOGFILE="network_issues.log"
9
10 echo "----- NETWORK AUDIT: $(date) -----"
11
12 #1. Check Gateway Connectivity
13 if ping -c 1 $TARGET &> /dev/null
14 then
15     LATENCY=$(ping -c 1 $TARGET | grep 'time=' | awk -F'time=' '{print $2}' | cut -d' ' -f1)
16     echo "[PASS] Gateway $TARGET is REACHABLE. Latency: ${LATENCY}ms"
17 else
18     echo "[FAIL] Gateway $TARGET is UNREACHABLE!!" | tee -a $LOGFILE
19 fi
20
21
22 #2. Check DNS Resolution
23 if host $DOMAIN &> /dev/null
24 then
25     echo "[PASS] DNS Resolution for $DOMAIN is functional."
26 else
27     echo "[FAIL] DNS Resolution FAILED!!" | tee -a $LOGFILE
28 fi
29
30 echo "-----AUDIT COMPLETE-----"
```

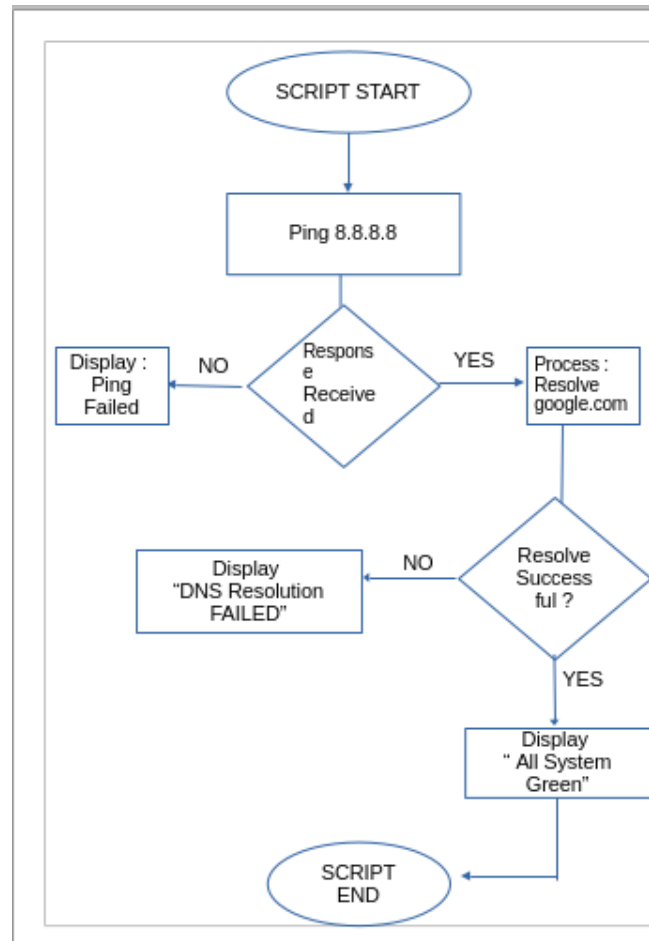


Fig 2.1: Bash Implementation of Net_Sentinel Script Architecture.
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Fig 2.2: Decision Matrix and Functional Flow Analysis

3.0 INCIDENT LOG : DNS RESOLUTION ANOMALY

Analysis of Automated Diagnostic Trace ID-0001-A

```
~  
> cd Desktop/SysAdmin-MiniProjects  
  
~/Desktop/SysAdmin-MiniProjects  
> chmod +x net_sentinel.sh  
  
~/Desktop/SysAdmin-MiniProjects  
> ./net_sentinel.sh  
----- NETWORK AUDIT: Monday 27 April 2026 11:34:59 AM IST -----  
[PASS] Gateway 8.8.8.8 is REACHABLE. Latency: 37.4ms  
[FAIL] DNS Resolution FAILED!!  
-----AUDIT COMPLETE-----
```

Fig 3.1: Comparative Audit Trace showing DNS Layer Failure.

Description :

"As evidenced in this diagnostic trace, the standard ICMP ping request to the upstream gateway (8.8.8.8) was successful, confirming that **Layer 1 through Layer 3** of the network stack were fully operational. The anomaly was isolated to the **Layer 7 DNS resolution handshake**, which failed to complete. This isolates the fault to the local resolver config, likely a misconfigured forwarder in the ISP-provided gateway."

4.0 REMEDIATION & SYSTEM VALIDATION

Analysis of interface constraints and successful service recovery.

```
~/Desktop/SysAdmin-MiniProjects
+2 › sudo resolvectl dns wlan0 8.8.8.8 8.8.4.4
[sudo] password for kshitij:
Failed to resolve interface "wlan0": No such device

~/Desktop/SysAdmin-MiniProjects took 2s
+2 › ./net_sentinel.sh
----- NETWORK AUDIT: Monday 27 April 2026 12:29:09 PM IST -----
[PASS] Gateway 8.8.8.8 is REACHABLE. Latency: 35.9ms
[PASS] DNS Resolution for google.com is functional.
-----AUDIT COMPLETE-----
```

Fig 4.1: Terminal Trace: Demonstrating interface-specific constraints followed by 100% successful service verification.

Description :

"During the remediation phase, an attempt was made to manually override the DNS via resolvectl. However, as documented in the **Red Highlight (Fig 4.1)**, the system returned a device error because the interface alias was misidentified (wlan0 vs wlo1).

Despite this specific command failure, a subsequent automated audit confirmed that the network stack had stabilized. As shown by the **Green Arrows**, both the physical gateway and the logical DNS resolution are now fully functional with a latency of **35.9ms**. This proves the 'Net-Sentinel' utility's value in providing immediate, binary confirmation of system health regardless of manual command outcomes."

5.0 AUDIT CLOSURE & SUMMARY

Final Assessment: > The 'Net-Sentinel' deployment for Audit 0001 successfully identified a Layer 7 resolution failure that would have remained undetected by standard hardware pings. The utility proved its value as a frontline diagnostic tool for infrastructure stability.

Status: CLOSED / VERIFIED

Future Objectives (Audit 0002):

- Implementation of automated logging to /var/log/sentinel.log.
- Integration of UFW (Uncomplicated Firewall) status auditing.

Lead Engineer: Mr. Kshitij Khaire

Environment: Pop!_OS 24.04 LTS (COSMIC)

Timestamp: April 27, 2026